# Byzantine-Resilient Stochastic Games for Federated Multi-Cloud Intrusion Detection

**Abstract**

Modern enterprises deploy applications across multiple cloud providers, creating complex security challenges that existing intrusion detection systems cannot address effectively. This fragmentation allows sophisticated attackers to exploit coordination gaps between cloud boundaries while defenders struggle with privacy constraints and diverse data characteristics across different deployment environments. We present FedGTD, a comprehensive framework integrating game theory with federated learning to enable collaborative threat detection across heterogeneous multi-cloud infrastructures while preserving data privacy and strategic defense capabilities. Our approach models cloud security as a strategic game between multiple defenders and adaptive adversaries, accounting for the extreme diversity of modern cloud environments—from industrial IoT networks running specialized protocols to containerized applications with ephemeral connections, to enterprise security operations centers processing millions of alerts daily. The framework introduces four key innovations: strategic game-theoretic modeling with formal equilibrium guarantees, Byzantine-resilient federated learning adapted for severe data imbalance and heterogeneous features, rigorous convergence analysis under adversarial conditions, and domain-specific adversarial training targeting protocol exploits, container vulnerabilities, and correlation attacks. Extensive evaluation on 15.9 million security events across three distinct domains demonstrates substantial improvements: 95.7–96.9% detection accuracy across Edge-IoT, container, and SOC environments, representing 4.8–8.8% gains over recent cloud security methods. The system retains 94.0% average performance under 20% Byzantine corruption (up to 95.3% against label flipping attacks), with 96.3% baseline accuracy on clean data, while reducing communication overhead by 50.7%. Deployments across major cloud providers validate real-time scalability to millions of events per second, establishing a practical foundation for industry-wide collaborative defense.

**Keywords:** Stochastic game theory; Federated learning; Martingale convergence; Cloud security and intrusion detection; Differential privacy and Byzantine resilience; Heterogeneous feature spaces.

# 1 Introduction

The exponential growth of cloud computing has fundamentally transformed the cybersecurity landscape, with 92% of enterprises now employing multi-cloud strategies spanning an average of 2.6 public cloud providers [? ]. This architectural evolution introduces critical security challenges that existing network intrusion detection systems (NIDS) fail to address adequately. Modern cloud environments encompass diverse deployment paradigms—from resource-constrained edge IoT devices running industrial protocols to containerized microservices orchestrated by Kubernetes, to enterprise Security Operations Centers (SOCs) processing millions of security events daily. Each domain exhibits unique characteristics that traditional security approaches cannot handle comprehensively.

The practical implications of this fragmentation manifest in several critical ways. Consider a financial services company operating across AWS, Azure, and Google Cloud: their customer data processing runs on AWS, their machine learning infrastructure on Google Cloud, and their compliance systems on Azure. A sophisticated attacker compromising edge devices in their AWS deployment could laterally move to container workloads in Google Cloud, eventually targeting their SOC systems in Azure—all while evading detection because each cloud provider's security tools cannot correlate events across boundaries. This scenario is not hypothetical; the 2024 cross-cloud supply chain attack affected 47 Fortune 500 companies precisely through this coordination gap [? ].

The fragmentation of security monitoring across cloud boundaries creates critical blind spots where sophisticated attacks can propagate undetected. Each cloud provider's proprietary security tools—AWS GuardDuty, Azure Sentinel, GCP Security Command Center—operate in isolation, unable to share threat intelligence due to privacy constraints and competitive considerations. Recent security incidents demonstrate this inadequacy: the 2021 SolarWinds supply chain compromise affected multiple cloud tenants through fragmented security monitoring [? ], while the 2023 Microsoft Exchange vulnerabilities (CVE-2023-23397) exploited coordination gaps between cloud security boundaries [? ]. Industry reports indicate that 68% of cloud breaches involve lateral movement across provider boundaries that evade detection due to fragmented visibility [? ].

The challenge is fundamentally compounded by the game-theoretic nature of cybersecurity, where intelligent adversaries continuously adapt their strategies in response to defensive measures [? ]. Traditional NIDS operate under the assumption of static attack patterns, failing to account for the strategic interaction between defenders and attackers. This limitation becomes particularly acute in federated environments where multiple organizations must collaborate on defense while maintaining data sovereignty and privacy, a challenge highlighted by recent Byzantine attacks on federated learning systems [? ].

## 1.1 Comprehensive Problem Analysis

The intersection of multi-cloud deployment, strategic adversaries, privacy requirements, and heterogeneous data characteristics creates a complex problem space that existing solutions fail to address comprehensively. We identify five fundamental challenges that motivate our research.

Modern cloud environments span fundamentally different security domains in terms of heterogeneous feature space complexity. Edge-IIoT systems process protocol-specific features including MQTT topics, Modbus function codes, and HTTP methods alongside flow statistics. Container orchestration platforms require analysis of ephemeral connections, namespace-isolated traffic, and service mesh patterns. SOC systems demand temporal aggregation of alert counts, entity correlation across more than 33 types such as DeviceId, AccountSid, and IpAddress, and incident triage prediction. Existing federated approaches assume uniform feature spaces, failing catastrophically under such heterogeneity.

Real-world cloud security data exhibits extreme and variable class imbalance that varies dramatically across domains. The ICS3D datasets exemplify this challenge where Edge-IIoT data contains 1.6 million normal flows versus 0.6 million attack instances representing a 2.67 to 1 ratio, container environments show 220 thousand benign versus 14 thousand attacks with a 15.7 to 1 ratio, while SOC environments face extreme imbalance with less than 0.8 percent true positive incidents among millions of alerts representing a 99 to 1 ratio. Standard techniques like SMOTE or class weighting fail when applied uniformly across such diverse imbalance patterns.

Data distributions vary not only across organizations but also across domains within the same organization, creating multi-dimensional non-IID distribution challenges. An enterprise's Edge-IIoT traffic from manufacturing floors differs fundamentally from their container traffic in cloud data centers and their SOC alert patterns. This multi-dimensional non-IID nature manifests as label distribution skew with different attack prevalence, feature distribution skew with varying protocol usage patterns, temporal distribution skew due to operational rhythms, and cross-domain distribution skew reflecting infrastructure differences.

Sophisticated attackers exploit the predictable behavior of existing defense systems through strategic adversarial adaptation. They can observe aggregate defensive responses, adapt attack patterns to evade detection thresholds, and coordinate multi-stage campaigns across cloud boundaries. The game-theoretic nature of this interaction requires formal modeling of strategic behavior under partial observability constraints.

Different domains have distinct privacy requirements creating privacy-preserving collaboration requirements. Edge-IIoT data may reveal industrial processes and operational schedules, container traffic exposes application architectures and deployment patterns, while SOC data contains sensitive entity relationships and security postures. Regulatory

frameworks including GDPR, HIPAA, and SOX prohibit direct data sharing, necessitating privacy-preserving collaboration mechanisms that maintain utility for security detection.

## 1.2 Mathematical Problem Formulation

We formalize the multi-cloud security challenge as a stochastic game where defenders must collaborate while preserving privacy and maintaining effectiveness against strategic adversaries. Let $\mathcal{D} = \{D_1, D_2, \ldots, D_K\}$ represent $K$ cloud organizations (defenders) and $\mathcal{A}$ represent the adversary coalition. Each defender $D_k$ operates in one or more security domains $d \in \{\text{Edge}, \text{Container}, \text{SOC}\}$ with local data $\mathcal{X}_k^{(d)} = \{(x_i^{(k,d)}, y_i^{(k,d)})\}_{i=1}^{n_k^{(d)}}$ where $x_i^{(k,d)} \in \mathbb{R}^{d_d}$ are $d_d$-dimensional feature vectors and $y_i^{(k,d)} \in \mathcal{Y}^{(d)}$ are domain-specific labels.

The fundamental challenge is to design a mechanism that optimizes the following objective:

$$\begin{aligned} \min_{\{\theta_k^{(d)}\}} \quad & \sum_d \sum_{k \in \mathcal{D}_d} w_k^{(d)} \mathcal{L}_k^{(d)}(\theta_k^{(d)}) \\ \text{subject to:} \quad & \mathcal{R}_{\text{privacy}}(\{\theta_k^{(d)}\}) \leq \epsilon_{\text{privacy}} \\ & \mathcal{R}_{\text{robustness}}(\{\theta_k^{(d)}\}, \pi_A) \leq \epsilon_{\text{robust}} \\ & \mathcal{R}_{\text{communication}}(\{\theta_k^{(d)}\}) \leq B_{\text{comm}} \end{aligned} \tag{1}$$

where $w_k^{(d)} = n_k^{(d)} / \sum_{j \in \mathcal{D}_d} n_j^{(d)}$ are domain-specific weights, $\mathcal{L}_k^{(d)}$ represents local loss functions adapted to domain-specific characteristics, $\pi_A$ is the adversarial strategy, and the regularization terms enforce privacy, robustness, and communication constraints.

We now describe each component in turn. The weighted sum in the objective aggregates the empirical risk across all $K$ defenders and all security domains, where the weights $w_k^{(d)}$ ensure that each client's contribution is proportional to its local dataset size within the domain, preventing clients with few samples from disproportionately biasing the global model. The local loss $\mathcal{L}_k^{(d)}(\theta_k^{(d)})$ is a cross-entropy function modified by class weights $w_c^{(d)} = \sqrt{n_{\text{total}}^{(d)} / (C_d \cdot n_c^{(d)})}$ that account for the severe class imbalance in each domain, where $C_d$ denotes the number of classes and $n_c^{(d)}$ is the sample count for class $c$.

The three constraints encode the essential design requirements of our framework. The privacy constraint $\mathcal{R}_{\text{privacy}}(\{\theta_k^{(d)}\}) \leq \epsilon_{\text{privacy}}$ bounds the cumulative information leakage about individual data points through the lens of $(\epsilon, \delta)$-differential privacy, ensuring that model updates cannot be reverse-engineered to reveal private training samples. The robustness constraint $\mathcal{R}_{\text{robustness}}(\{\theta_k^{(d)}\}, \pi_A) \leq \epsilon_{\text{robust}}$ guarantees that the aggregated global model deviates from the honest aggregate by no more than $\epsilon_{\text{robust}}$ in $\ell_2$ norm, even when a fraction $f$ of participating clients submit adversarial updates under the adversary strategy $\pi_A$. Because this constraint depends on the adversarial strategy, the optimizer must find parameters resilient to the worst-case adversary within the allowed corruption budget,

which introduces a minimax structure that our Nash equilibrium analysis resolves. The communication constraint $\mathcal{R}_{\text{communication}}(\{\theta_k^{(d)}\}) \leq B_{\text{comm}}$ limits the total volume of model parameters exchanged per federated round, which is critical for practical deployment at scale.

## 1.3 Research Contributions

This paper addresses these fundamental challenges through a novel integration of stochastic game theory, federated learning, and martingale analysis, providing both theoretical guarantees and practical effectiveness. Our contributions advance the state-of-the-art across multiple dimensions.

First, we present a unified stochastic game-theoretic framework that formulates multi-cloud security as a continuous-time stochastic differential game capturing the strategic interaction between multiple defenders and adaptive adversaries under partial observability. Our framework explicitly models heterogeneous feature spaces, extreme class imbalance, and temporal dynamics through jump-diffusion processes, providing the first comprehensive game-theoretic model for federated security in heterogeneous environments.

Second, we develop a Byzantine-resilient heterogeneous federated protocol that maintains Byzantine resilience despite extreme heterogeneity across domains. Our approach combines gradient similarity screening adapted for varying feature spaces through projection to common subspaces, domain-specific trimmed-mean aggregation with adaptive trim ratios based on data quality and imbalance severity, reputation tracking that accounts for natural performance variations across domains, and differential privacy calibration that preserves minority class signals while meeting regulatory requirements.

Third, we provide the first convergence proof for federated game-theoretic learning under heterogeneous conditions through rigorous martingale-based analysis. Our analysis incorporates domain-specific learning rates adjusted for imbalance severity, multi-scale temporal kernels capturing attack campaigns and operational patterns, cross-domain variance bounds accounting for feature space differences, and explicit convergence rates to epsilon-Nash equilibrium despite non-convex objectives and temporal non-stationarity.

Fourth, we introduce domain-adaptive adversarial training with perturbation strategies tailored to each security domain, including protocol-aware attacks for Edge-IIoT such as MQTT topic manipulation and Modbus function code exploits, flow-based perturbations for containers including inter-arrival time modifications and packet size manipulations, and entity-correlation attacks for SOC systems involving temporal pattern disruption and entity masquerading. This domain-specific approach significantly improves robustness compared to generic adversarial training.

Fifth, we conduct comprehensive empirical validation on the complete Integrated Cloud Security 3Datasets, demonstrating 95.7 percent accuracy on 2.2 million Edge-IIoT flows

detecting all 14 attack families, 96.3 percent accuracy on container attacks including specific CVEs such as CVE-2019-5736 and CVE-2021-25741, 96.9 percent triage accuracy on over 13 million SOC events across 33 entity types, successful cross-domain transfer learning with 18 to 24 percent performance improvement, and sub-second per-round convergence enabling real-time deployment.

## 1.4 Notation and Symbols

Table **??** summarizes the key mathematical notation used throughout this paper. This comprehensive notation table ensures clarity and consistency in our mathematical formulations, particularly critical given the heterogeneous nature of our framework spanning multiple security domains with distinct characteristics.

Table 1: Notation Summary

| **Symbol** | **Description** |
|---|---|
| *Players and Domains* | |
| $\mathcal{D} = \{D_1, \ldots, D_K\}$ | Set of $K$ defender organizations |
| $\mathcal{A} = \{A_1, \ldots, A_M\}$ | Adversary coalition with $M$ attackers |
| $\mathcal{D}_{\text{Edge}}, \mathcal{D}_{\text{Container}}, \mathcal{D}_{\text{SOC}}$ | Domain-specific defender subsets |
| $K$ | Total number of defender organizations |
| $d \in \{\text{Edge}, \text{Container}, \text{SOC}\}$ | Security domain identifier |
| *Data and Features* | |
| $\mathcal{X}_k^{(d)}$ | Local dataset for client $k$ in domain $d$ |
| $x_i^{(k,d)} \in \mathbb{R}^{d_d}$ | Feature vector with $d_d$ dimensions |
| $y_i^{(k,d)} \in \mathcal{Y}^{(d)}$ | Label from domain-specific label space |
| $n_k^{(d)}$ | Number of samples for client $k$ in domain $d$ |
| $n = \sum_{d,k} n_k^{(d)}$ | Total number of samples across all clients |
| *Class Imbalance* | |
| $\rho_d$ | Imbalance ratio for domain $d$ |
| $w_c^{(d)} = \sqrt{n_{\text{total}}^{(d)}/(C_d \cdot n_c^{(d)})}$ | Class weight for class $c$ in domain $d$ |
| $C_d$ | Number of classes in domain $d$ |
| $n_c^{(d)}$ | Number of samples in class $c$ for domain $d$ |
| $\omega_d = 1/\rho_d$ | Domain weight inversely proportional to imbalance |
| *Model Parameters* | |
| $\theta_k^{(d)}$ | Model parameters for client $k$ in domain $d$ |
| $\theta_k^{(d,*)}$ | Nash equilibrium parameters |
| $\theta_{\text{global}}^{(d)}$ | Global model for domain $d$ |

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*Table 1 (continued)*

| Symbol | Description |
|---|---|
| $g_k^{(d,t)}$ | Gradient for client $k$ in domain $d$ at round $t$ |
| $\tilde{g}_k^{(d,t)}$ | Clipped gradient |
| *Learning Dynamics* | |
| $\eta_d^{(t)} = \eta_0 \rho_d^{1/2}/(t+1)^{2/3}$ | Domain-adaptive learning rate |
| $\eta_0$ | Base learning rate |
| $T$ | Total number of federated rounds |
| $E$ | Number of local epochs per round |
| *Privacy Parameters* | |
| $\epsilon_d$ | Privacy budget (epsilon) for domain $d$ |
| $\delta_d$ | Privacy parameter (delta) for domain $d$ |
| $\sigma_d$ | Noise standard deviation for domain $d$ |
| $C_d$ | Gradient clipping norm for domain $d$ |
| *Byzantine Resilience* | |
| $\mathcal{B}_d$ | Set of detected Byzantine clients in domain $d$ |
| $f$ or $f_d$ | Number/fraction of Byzantine clients |
| $\alpha_d$ | Trim ratio for domain $d$ |
| $r_k^{(d,t)}$ | Reputation score for client $k$ |
| $\tau_d$ | Byzantine detection threshold for domain $d$ |
| *Game-Theoretic Components* | |
| $\mathcal{S}$ | Global state space |
| $\mathcal{S}_{\text{het}}$ | Heterogeneous state space |
| $s_t$ | System state at time $t$ |
| $\pi_D^*, \pi_A^*$ | Nash equilibrium strategies |
| $\Delta_{\text{Nash}}^{(t)}$ | Nash gap at round $t$ |
| $J_A$ | Adversary objective function |
| $u_k^{(d)}$ | Payoff function for defender $k$ in domain $d$ |
| *Stochastic Processes* | |
| $W_t^{(d)}$ | Brownian motion for domain $d$ |
| $N_i^{(d)}(dt, dz)$ | Poisson random measure for attack type $i$ |
| $\mathcal{F}_t$ | Filtration at time $t$ |
| $\mu_d, \sigma_d$ | Drift and diffusion coefficients |
| *Common Mathematical Notation* | |
| $\mathbb{R}$ | Real numbers |
| $\mathbb{E}[\cdot]$ | Expectation operator |
| $\|\cdot\|_2$ | Euclidean (L2) norm |

*Continued on next page*

*Table 1 (continued)*

| Symbol | Description |
|---|---|
| $\mathcal{N}(\mu, \sigma^2)$ | Gaussian distribution |
| a.s. | Almost surely (probability 1) |

# 2 Related Work

The intersection of game theory, federated learning, and cybersecurity represents an emerging research area with significant theoretical and practical challenges. We review the relevant literature across four key dimensions that inform our approach, with additional details provided in Appendix A.

## 2.1 Game-Theoretic Approaches to Network Security

The application of game theory to cybersecurity has evolved significantly since the foundational work of Alpcan and Başar [? ], who established the mathematical framework for modeling network defense as a two-player zero-sum game. Recent advances by Zhu and Başar [? ] incorporate dynamic aspects through differential games, establishing Hamilton-Jacobi-Bellman equations for optimal defense strategies. However, their computational approach scales exponentially with state space dimension, limiting practical applications to small-scale systems. Multi-criteria evaluation frameworks have also been applied to security assessment in emerging technologies, including spherical fuzzy methods for analyzing privacy risks in metaverse environments [? ], demonstrating that structured decision-making under uncertainty remains central to security analysis across domains.

Manshaei et al. [? ] provided a comprehensive survey of game-theoretic approaches to network security, identifying key challenges including incomplete information, bounded rationality, and mechanism design for cooperative defense. However, their analysis focuses primarily on traditional network environments and does not address the unique challenges of cloud-native architectures, containerized applications, or IoT device integration (see Appendix A.1 for extended discussion).

## 2.2 Federated Learning for Network Security

Federated learning, introduced by McMahan et al. [? ], enables collaborative model training without centralizing sensitive data. The seminal FedAvg algorithm demonstrated that distributed gradient descent could achieve comparable performance to centralized training while preserving data locality. However, the original formulation assumes IID data distributions and benign participants, assumptions that rarely hold in security applications.

Li et al. [? ] identified key challenges in federated learning: statistical heterogeneity (non-IID data), systems heterogeneity (device capabilities), and privacy concerns. Their FedProx algorithm addresses non-IID distributions through proximal terms in local objectives, but the approach assumes mild heterogeneity and fails under the extreme variations observed in multi-cloud security data.

Recent work by Al-Hawawreh et al. [? ] applied federated learning to IoT intrusion detection, showing improved performance over centralized approaches. Related forensic analysis methods for IoT communications [? ] underscore the need for intelligent, multi-modal threat detection across diverse network environments, a challenge that federated approaches are particularly well suited to address. Advanced federated learning approaches for cloud security have emerged recently, including CloudFL [? ] achieving 92.1% accuracy on multi-cloud datasets but without game-theoretic considerations, and AdaptiveFL-Container [? ] with 89.7% detection rates for container security (detailed comparison in Appendix A.2).

## 2.3 Byzantine-Robust Federated Learning

The vulnerability of federated learning to Byzantine attacks was first demonstrated by Blanchard et al. [? ], who showed that a single malicious client can arbitrarily manipulate the global model. Yin et al. [? ] introduced coordinate-wise median and trimmed mean aggregation, achieving breakdown points near 50% against arbitrary corruptions. However, their analysis assumes homogeneous data distributions and may fail when natural heterogeneity across domains mimics Byzantine behavior.

Recent advances include the BALANCE algorithm by Zhang et al. [? ], which enables each client to use its local model as a similarity reference to detect malicious updates. Zhou et al. [? ] introduced RobustFL with 94.2% accuracy under 30% Byzantine corruption (extended analysis in Appendix A.3).

## 2.4 Critical Gaps and Limitations

The existing literature exhibits three fundamental inadequacies that our work addresses. First, the *architectural mismatch*: prior federated learning approaches [? ? ] implicitly assume clients operate in similar computational environments with comparable data characteristics. This assumption breaks catastrophically in multi-cloud scenarios where feature spaces may be entirely disjoint.

Second, the *imbalance severity gap*: while recent work acknowledges class imbalance [? ? ], no existing approach handles the extreme ratios observed in production security systems. The difference between handling 3:1 imbalance (typical in prior work) versus 99:1 imbalance (SOC reality) is not merely quantitative but qualitative.

Third, the *adversarial model disconnect*: Byzantine-robust federated learning [? ? ] focuses on random or adversarial parameter corruption, but real attackers in security scenarios are *strategic*—they observe defensive responses and adaptively modify attack patterns to maximize infiltration while minimizing detection probability. Additional gaps are discussed in Appendix A.4.

# 3 Problem Formulation

## 3.1 System Model and Architecture

We consider a heterogeneous multi-cloud federation consisting of $K$ organizations (defenders) $\mathcal{D} = \{D_1, D_2, \ldots, D_K\}$ operating across diverse cloud infrastructures with distinct security domains. Each defender $D_k$ manages one or more deployment types characterized by fundamentally different data characteristics, threat models, and operational requirements.

**Definition 3.1** (Heterogeneous Multi-Cloud Security Domains)**.** The federation encompasses three primary security domains with distinct mathematical properties:

1. Edge-IIoT Domain $\mathcal{D}_{\text{Edge}} \subseteq \mathcal{D}$: Organizations managing industrial IoT deployments with protocol-specific traffic characterized by high-dimensional feature vectors $x_i^{(\text{Edge})} \in \mathbb{R}^{d_{\text{Edge}}}$ where $d_{\text{Edge}} \in [60, 140]$ depending on protocol diversity (MQTT, Modbus, CoAP, HTTP, DNS, TCP/UDP).

2. Container Domain $\mathcal{D}_{\text{Container}} \subseteq \mathcal{D}$: Organizations operating Kubernetes clusters and microservices with flow-based features $x_i^{(\text{Container})} \in \mathbb{R}^{87}$ capturing directional statistics, inter-arrival times, packet characteristics, and service mesh patterns.

3. SOC Domain $\mathcal{D}_{\text{SOC}} \subseteq \mathcal{D}$: Enterprise security operations centers processing multi-entity incidents with mixed-type features $x_i^{(\text{SOC})} \in \mathbb{R}^{46}$ including categorical entities (33 types), temporal aggregates, and incident metadata.

The heterogeneous nature of these domains introduces fundamental mathematical challenges that require careful treatment. Different security domains operate in largely disjoint feature spaces with minimal semantic overlap, where the common feature space contains only basic flow identifiers (timestamps, IP addresses, ports, protocols) constituting less than 10% of total features, while domain-specific features remain disjoint.

**Definition 3.2** (Multi-Cloud Security State Space)**.** The global security state at time $t$ incorporates domain-specific components and their interactions:

$$s_t = \Big( \{s_t^{(k,\text{Edge})}\}_{k \in \mathcal{D}_{\text{Edge}}}, \{s_t^{(k,\text{Container})}\}_{k \in \mathcal{D}_{\text{Container}}}, \\ \{s_t^{(k,\text{SOC})}\}_{k \in \mathcal{D}_{\text{SOC}}}, s_t^{(\mathcal{A})} \Big) \in \mathcal{S} \tag{2}$$

where each component captures domain-specific security posture, model parameters, reputation scores, and adversarial state information.

## 3.2 Comprehensive Threat Model

The threat model must account for the diverse attack surfaces and capabilities across heterogeneous domains while modeling strategic adversarial behavior. Risk management approaches for cyber-physical systems, such as the security-by-design framework proposed for autonomous vehicles [**?** ], inform our multi-domain threat modeling by highlighting the importance of integrating security considerations at the architectural level rather than treating them as post-hoc additions.

**Definition 3.3** (Multi-Domain Adversary Coalition)**.** The adversary coalition

$$\mathcal{A} = \{A_1, A_2, \ldots, A_M\}$$

possesses domain-specific capabilities and strategic coordination across three primary attack vectors.

The Edge-IIoT attack arsenal includes protocol exploitation through MQTT topic fuzzing, Modbus function code manipulation, and CoAP resource discovery attacks. Volume-based attacks include DDoS variants targeting industrial control systems with timing constraints. Persistence mechanisms involve ransomware deployment and backdoor installation in IoT firmware. Data integrity attacks operate through sensor value tampering, command injection, and false data injection.

Container attack vectors encompass vulnerability exploitation including CVE-2019-5736 for container escape and CVE-2021-25741 for privilege escalation. Application-layer attacks include Node-RED RCE and dashboard exploits such as CVE-2021-43798. Supply chain attacks involve container image poisoning and dependency confusion attacks. Orchestration abuse targets Kubernetes API server exploitation and admission controller bypass.

SOC evasion techniques utilize alert fatigue exploitation through high-volume false positive generation to mask real attacks. Entity masquerading spoofs legitimate device and user behavioral patterns. Temporal correlation breaking employs timing attacks designed to evade detection windows. Analyst workflow exploitation mimics normal user behavior during shift changes.

Cross-domain coordination enables multi-stage campaigns following patterns from IoT compromise to container lateral movement to SOC evasion. Information fusion attacks combine intelligence across domains to plan sophisticated attacks. Adaptive strategy evolution learns from defensive responses across all domains.

### 3.2.1 Threat Model Assumptions and Boundaries

We make the following explicit assumptions regarding adversary capabilities and constraints, which underpin the robustness guarantees established in Theorem **??**. First, in each domain $d$, the fraction $f_d$ of Byzantine clients satisfies $f_d < \alpha_d/2$, where $\alpha_d$ is the domain-specific trim ratio. Second, Byzantine clients may submit arbitrary gradient updates but cannot observe other clients' private gradients before submitting their own (the honest-but-curious server assumption). Third, honest clients' gradients satisfy a bounded variance condition $\mathbb{E}[\|g_k^{(d,t)} - \nabla\mathcal{L}^{(d)}(\theta^{(d,t)})\|^2] \leq \sigma_d^2$ for a domain-specific variance bound $\sigma_d^2$. Fourth, Byzantine clients may collude and coordinate their attack strategies, but the total number of colluding clients per domain remains below the Byzantine threshold. When these assumptions are violated, for instance when Byzantine fractions exceed $\alpha_d/2$, performance degrades gracefully rather than failing catastrophically; Section 6.3 reports empirical results under both assumption-satisfying and assumption-exceeding scenarios.

The adversary's objective function captures both domain-specific goals and cross-domain synergies:

$$\begin{aligned} J_A = \mathbb{E}\Bigg[ \int_0^T \Bigg( & \sum_{k\in\mathcal{D}_{\text{Edge}}} \alpha_{\text{Edge}} \cdot I_{\text{Edge}}^{(k)}(t) \\ & + \sum_{k\in\mathcal{D}_{\text{Container}}} \alpha_{\text{Container}} \cdot I_{\text{Container}}^{(k)}(t) + \sum_{k\in\mathcal{D}_{\text{SOC}}} \alpha_{\text{SOC}} \cdot I_{\text{SOC}}^{(k)}(t) \\ & + \gamma \cdot \Psi_{\text{CrossDomain}}(t) - \beta \cdot D(t) - \lambda_A \|r_A(t)\|_2^2 \Bigg) dt \Bigg] \end{aligned} \tag{3}$$

where $I_d^{(k)}(t)$ represents infiltration success in domain $d$ for organization $k$, $\Psi_{\text{CrossDomain}}(t)$ captures cross-domain attack synergies, $D(t)$ is the overall detection probability, $r_A(t)$ represents adversarial resource consumption, and $\alpha_d, \gamma, \beta, \lambda_A$ are weight parameters reflecting attack priorities and resource constraints.

## 3.3 Federated Detection with Heterogeneous Features

The core challenge lies in designing a federated learning mechanism that can handle the extreme heterogeneity across domains while maintaining privacy and robustness guarantees.

**Definition 3.4** (Domain-Adaptive Federated Objective)**.** The federated learning objective must accommodate heterogeneous domains, extreme class imbalance, and adversarial

threats:

$$\min_{\{\theta_k^{(d)}\}} \mathcal{L}_{\text{fed}} = \sum_{d\in\{\text{E,C,S}\}} \sum_{k\in\mathcal{D}_d} \frac{n_k^{(d)}}{n} \cdot \omega_d^{-1} \cdot \mathcal{L}_k^{(d)}(\theta_k^{(d)}) \tag{4}$$
$$+ \lambda_1 \mathcal{R}_{\text{privacy}}(\{\theta_k^{(d)}\}) + \lambda_2 \mathcal{R}_{\text{balance}}(\{\theta_k^{(d)}\}, \{\rho_d\})$$
$$+ \lambda_3 \mathcal{R}_{\text{temporal}}(\{\theta_k^{(d)}\}, t) + \lambda_4 \mathcal{R}_{\text{strategic}}(\{\theta_k^{(d)}\}, \pi_A)$$

where $n_k^{(d)}$ is the number of samples for client $k$ in domain $d$, $n = \sum_{d,k} n_k^{(d)}$ is the total sample count, $\omega_d = \sqrt{\rho_d}$ is the imbalance weighting factor where $\rho_d$ is the imbalance ratio for domain $d$, and the various regularization terms enforce privacy, balance, temporal consistency, and strategic considerations.

Each domain exhibits severe but different class imbalance patterns. Edge-IIoT shows approximately 2.67:1 normal to attack ratio, Container environments demonstrate 15.7:1 benign to attack ratio, while SOC environments face extreme 99:1 imbalance between false/benign positives and true positives. Moreover, within-domain attack distributions are highly skewed, with rare attack types comprising less than 0.1% of samples in each domain. These patterns necessitate our domain-adaptive weighting strategy to ensure effective learning across all security domains despite the extreme data heterogeneity.

## 3.4 Comprehensive Data Characteristics Analysis

To properly formulate the mathematical framework, we must understand the precise characteristics of the Integrated Cloud Security 3Datasets (ICS3D) that drive our approach.

**Definition 3.5** (ICS3D Dataset Specifications)**.** The Integrated Cloud Security 3Datasets comprise three interconnected components representing the full spectrum of modern cloud security challenges.

The Edge-IIoT component contains 2,219,201 network flows with 60 to 140 dimensions depending on protocol diversity, spanning 14 distinct attack families including DDoS, MITM, Ransomware, and Backdoor attacks across MQTT, Modbus, HTTP, DNS, TCP, and UDP protocols with protocol-specific fields. The class distribution shows Normal traffic at 72.1 percent with attacks distributed across families, collected over multi-day spans with diurnal patterns.

The Container component includes 234,560 microservice flows with 87 CICFlowMeter-style attributes, covering 11 CVE-based exploits and application attacks in Kubernetes-orchestrated containers with service mesh infrastructure. The class distribution shows Benign traffic at 94 percent and attacks at 6 percent with CVE-specific patterns, characterized by ephemeral connections, namespace isolation, and overlay networks.

The SOC component encompasses more than 13,449,329 evidence rows, 1,311,743 alerts, and 927,119 incidents with 46 mixed-type attributes including entity IDs, temporal

aggregates, and metadata across 33 entity categories such as DeviceId, AccountSid, and IpAddress. The incident classification shows False Positive at 96.0 percent, Benign Positive at 3.2 percent, and True Positive at 0.8 percent with multi-window aggregations over 1 hour, 24 hour, and 7 day periods reflecting real SOC analyst workflows.

Each domain exhibits distinct temporal characteristics that must be modeled appropriately. These patterns can be captured through multi-scale temporal features:

$$\begin{aligned}\phi_{\text{temporal}}(x_t) = \big[\, &\text{win}_{1s}(t),\ \text{win}_{10s}(t),\ \ldots,\ \text{win}_{1h}(t), \\ &\sin\!\left(\tfrac{2\pi t}{T_{\text{day}}}\right),\ \cos\!\left(\tfrac{2\pi t}{T_{\text{day}}}\right)\big]\end{aligned} \tag{5}$$

where $\text{win}_\tau(t)$ represents aggregated statistics over window $\tau$ and the sinusoidal terms capture diurnal patterns.

## 3.5 Privacy and Regulatory Requirements

The federated learning framework must satisfy stringent privacy requirements that vary across domains and jurisdictions.

**Definition 3.6** (Domain-Specific Privacy Requirements)**.** Each domain imposes distinct privacy constraints that reflect the sensitivity of the underlying data and regulatory requirements.

Edge-IIoT deployments require industrial process confidentiality, operational schedule privacy, and device capability concealment with privacy budget constraints of epsilon not exceeding 2.5 and delta not exceeding 10 to the power of negative 5.

Container environments demand application architecture privacy, deployment pattern confidentiality, and performance characteristic concealment with stricter requirements where epsilon does not exceed 2.0 and delta does not exceed 10 to the power of negative 6.

SOC operations necessitate entity relationship privacy, security posture confidentiality, and incident response capability concealment with the most stringent constraints where epsilon does not exceed 1.8 and delta does not exceed 10 to the power of negative 7, reflecting the highly sensitive nature of security operations data.

These privacy budgets are carefully calibrated to meet regulatory requirements while maintaining sufficient utility for effective intrusion detection. The domain-specific calibration reflects the varying sensitivity levels and regulatory constraints across different security domains, as detailed further in Appendix B.

# 4 Methodology

## 4.1 Stochastic Game-Theoretic Framework

We develop a comprehensive mathematical framework that models the multi-cloud security scenario as a continuous-time stochastic differential game, explicitly accounting for the heterogeneous nature of modern cloud environments.

### 4.1.1 Intuitive Framework Overview

Before presenting the formal mathematical framework, we provide an intuitive understanding of our approach. Imagine three organizations: a manufacturing company monitoring industrial IoT devices, a financial services firm running containerized trading applications, and a healthcare provider operating a security operations center. Each faces sophisticated cyber attackers, but they cannot directly share their security data due to privacy regulations and competitive concerns.

Our framework models this scenario as a strategic game where defenders (the three organizations) must coordinate their defense strategies while attackers adapt their tactics based on observed defensive responses. The key insight is that attackers cannot simultaneously optimize against all three diverse environments—an exploit effective against IoT protocols may fail in containerized environments, while SOC evasion techniques may leave detectable signatures in network flows.

The federated learning component enables these organizations to collaboratively train intrusion detection models without sharing raw data. However, unlike standard federated learning, our approach explicitly accounts for the fact that these organizations operate in fundamentally different environments: the IoT manufacturer's data consists of structured protocol messages with strict timing constraints, the financial firm's data comprises millions of ephemeral HTTP connections with complex payload patterns, and the healthcare provider's data includes entity relationships and temporal correlations across diverse alert types.

The game-theoretic component ensures that the collaborative defense strategies converge to a Nash equilibrium—a stable configuration where no single defender benefits from unilaterally changing their strategy, and attackers cannot improve their infiltration success without triggering detection. This equilibrium provides formal guarantees that our federated defense remains effective even as attackers adaptively modify their tactics.

### 4.1.2 Heterogeneous Multi-Domain Security Game

**Definition 4.1** (ICS3D Federated Security Game)**.** The heterogeneous security game accounting for domain diversity is defined as:

$$\mathcal{G}_{\text{ICS3D}} = (\mathcal{N}, \S_{\text{het}}, \mathcal{A}_{\text{het}}, T_{\text{het}}, R_{\text{het}}, \gamma, \mathcal{F}, P, \Theta) \tag{6}$$

where $\mathcal{N} = \mathcal{D}_{\text{Edge}} \cup \mathcal{D}_{\text{Container}} \cup \mathcal{D}_{\text{SOC}} \cup \mathcal{A}$ represents all players with domain-specific roles, $\S_{\text{het}} = \S_{\text{Edge}}^{|\mathcal{D}_E|} \times \S_{\text{Container}}^{|\mathcal{D}_C|} \times \S_{\text{SOC}}^{|\mathcal{D}_S|} \times \S_A$ is the heterogeneous state space, $\mathcal{A}_{\text{het}} = \prod_d \mathcal{A}_d^{|\mathcal{D}_d|} \times \mathcal{A}_A$ includes domain-specific defender actions and adversarial actions, $T_{\text{het}} : \S_{\text{het}} \times \mathcal{A}_{\text{het}} \times \S_{\text{het}} \to [0,1]$ models cross-domain state transitions including attack propagation, $R_{\text{het}} : \S_{\text{het}} \times \mathcal{A}_{\text{het}} \to \mathbb{R}^{|\mathcal{N}|}$ incorporates imbalance-weighted rewards, $\gamma \in (0,1)$ is the discount factor, $\mathcal{F} = \{\mathcal{F}_t\}_{t \geq 0}$ represents the filtration capturing information flow across domains, $P$ is the probability measure on the canonical space, and $\Theta$ captures temporal correlation structure across domains.

### 4.1.3 Continuous-Time Dynamics with Domain-Specific Characteristics

The system evolution must capture the unique dynamics of each domain while modeling their interactions through coupled stochastic differential equations:

$$\begin{aligned} ds_t^{(\text{Edge})} &= \mu_{\text{Edge}}(s_t, a_t, t)dt + \sigma_{\text{Edge}}(s_t, a_t, t)dW_t^{(1)} \\ &\quad + \sum_{i=1}^{14} \gamma_i^{(\text{Edge})} \int_{\mathbb{R}} z N_i^{(\text{Edge})}(dt, dz) \end{aligned} \tag{7}$$

$$\begin{aligned} ds_t^{(\text{Container})} &= \mu_{\text{Container}}(s_t, a_t, t)dt + \sigma_{\text{Container}}(s_t, a_t, t)dW_t^{(2)} \\ &\quad + \sum_{j=1}^{11} \gamma_j^{(\text{Container})} \int_{\mathbb{R}} z N_j^{(\text{Container})}(dt, dz) \end{aligned} \tag{8}$$

$$\begin{aligned} ds_t^{(\text{SOC})} &= \mu_{\text{SOC}}(s_t, a_t, t)dt + \sigma_{\text{SOC}}(s_t, a_t, t)dW_t^{(3)} \\ &\quad + \sum_{e=1}^{33} \gamma_e^{(\text{SOC})} \int_{\mathcal{E}} \eta(e) N_e^{(\text{SOC})}(dt, de) \end{aligned} \tag{9}$$

where $W_t^{(d)}$ are independent Brownian motions for each domain $d$, $N_i^{(\text{Edge})}(dt, dz)$ are Poisson random measures for 14 Edge-IIoT attack families, $N_j^{(\text{Container})}(dt, dz)$ model the 11 container exploit scenarios, $N_e^{(\text{SOC})}(dt, de)$ capture entity-specific alert patterns across 33 entity types, and $\gamma_\cdot^{(d)}$ are domain-specific jump intensity functions.

**Theorem 4.2** (Existence and Uniqueness of Heterogeneous Solution)**.** Under domain-specific Lipschitz conditions, bounded cross-domain coupling, and integrable jump measures, the coupled system of stochastic differential equations (**??**)–(**??**) admits a unique strong solution for any initial condition $s_0 \in \S_{\text{het}}$.

*Proof.* The proof follows standard techniques for existence and uniqueness of solutions to stochastic differential equations with jumps. The domain-specific Lipschitz conditions ensure local Lipschitz continuity of the drift and diffusion coefficients, while bounded cross-domain coupling limits the growth of these coefficients. The integrability of jump measures ensures that the compensated Poisson random measures are well-defined martingales. By applying Picard iteration in the space of càdlàg processes with appropriate norms, we establish existence and uniqueness of the strong solution. Detailed proof is provided in Appendix C.1. □

### 4.1.4 Nash Equilibrium Under Extreme Imbalance

The game-theoretic analysis must account for the extreme class imbalance observed across domains, which fundamentally affects the payoff structure and equilibrium characterization.

**Definition 4.3** (Imbalance-Adjusted Payoff Functions)**.** For defender $k$ in domain $d$, the payoff function incorporates class imbalance through weighted utilities:

$$\begin{aligned} u_k^{(d)}(s,a) = \sum_{c=1}^{C_d} w_c^{(d)} \cdot \big(&\alpha^{(d)} \cdot \text{DetectionReward}_c(s,a) \\ &-\beta^{(d)} \cdot \text{FalsePositiveCost}_c(s,a) - \gamma^{(d)} \cdot \text{ResourceCost}(a)\big) \end{aligned} \tag{10}$$

where $w_c^{(d)} = \sqrt{n_{\text{total}}^{(d)}/(C_d \cdot n_c^{(d)})}$ are inverse square-root frequency weights for class $c$ in domain $d$, and $C_d$ is the number of classes in domain $d$.

**Theorem 4.4** (Existence of Imbalance-Robust Nash Equilibrium)**.** For the ICS3D federated security game with imbalance ratios $\rho_d \in [2.67, 99]$ across domains, a mixed strategy Nash equilibrium $\pi^* = (\pi_D^*, \pi_A^*)$ exists under appropriate regularity conditions, and satisfies the imbalance-robustness bound:

$$\max_d \sqrt{\rho_d} \cdot \|\pi_d^* - \pi_d^{\text{uniform}}\|_1 \leq C\sqrt{\log K} \tag{11}$$

where $C$ is a universal constant and $K$ is the total number of defenders.

*Proof sketch.* The existence of Nash equilibrium follows from Kakutani's fixed-point theorem applied to the best-response correspondence. The key technical challenge is proving that the imbalance-weighted payoffs satisfy the required continuity and convexity properties. The robustness bound follows from concentration inequalities for the weighted empirical distributions combined with uniform convergence results for the payoff functions. Complete proof is provided in Appendix C.2. □

## 4.2 Byzantine-Resilient Federated Protocol for Heterogeneous Domains

The federated learning protocol must maintain robustness against Byzantine attacks while handling the extreme heterogeneity across domains through our novel secure aggregation approach.

**Algorithm 1** ICS3D Byzantine-Resilient Secure Aggregation

**Require:** Domain models $\{\theta_k^{(d)}\}$, imbalance ratios $\{\rho_d\}$, Byzantine bound $f$
**Ensure:** Robust global models $\{\theta_{\text{global}}^{(d)}\}$

**Domain-Specific Gradient Processing:**
**for** each domain $d \in \{\text{Edge}, \text{Container}, \text{SOC}\}$ **do**
  **for** each client $k \in \mathcal{D}_d$ **do**
    Compute imbalance-weighted gradient:
    $g_k^{(d)} \leftarrow \sum_{c=1}^{C_d} w_c^{(d)} \nabla \mathcal{L}_c^{(k,d)}(\theta_k^{(d)})$
    Apply domain-specific clipping:
    $\tilde{g}_k^{(d)} \leftarrow \text{clip}(g_k^{(d)}, C_d \cdot \rho_d^{-1/2})$
    Generate privacy mask: $r_k^{(d)} \sim \text{Uniform}(-R_d, R_d)^{|\theta|}$
    Compute commitment: $c_k^{(d)} \leftarrow \text{Hash}(\tilde{g}_k^{(d)} \| r_k^{(d)})$
  **end for**
**end for**
**Cross-Domain Byzantine Detection:**
**for** each domain $d$ **do**
  Project gradients to common subspace: $\hat{g}_k^{(d)} \leftarrow P_d \tilde{g}_k^{(d)}$
  Compute pairwise similarities: $S_{ij}^{(d)} = \cos(\hat{g}_i^{(d)}, \hat{g}_j^{(d)})$
  Identify outliers using domain-specific thresholds:
  $\mathcal{B}_d \leftarrow \{k : \text{median}(\{S_{kj}^{(d)}\}_j) < \tau_d\}$
**end for**
**Hierarchical Robust Aggregation:**
**for** each domain $d$ **do**
  Apply trimmed mean with domain-specific trim ratio:
  $\theta_d^{\text{agg}} \leftarrow \text{TrimmedMean}(\{\tilde{g}_k^{(d)} : k \notin \mathcal{B}_d\}, \alpha_d)$
  Add calibrated differential privacy noise:
  $\theta_d^{\text{private}} \leftarrow \theta_d^{\text{agg}} + \mathcal{N}(0, \sigma_d^2 I)$
**end for**
**Cross-Domain Model Synthesis:**
Compute domain-weighted global update considering imbalance:
$\theta_{\text{global}} \leftarrow \sum_d \frac{|\mathcal{D}_d| \cdot \rho_d^{-1/2}}{\sum_{d'} |\mathcal{D}_{d'}| \cdot \rho_{d'}^{-1/2}} \theta_d^{\text{private}}$
**return** $\{\theta_{\text{global}}^{(d)}\}_d$

**Theorem 4.5** (Privacy and Robustness Guarantees)**.** Algorithm **??** achieves simultaneous privacy and robustness guarantees. For privacy, the algorithm satisfies $(\epsilon_d, \delta_d)$-differential

privacy per domain with

$$\epsilon_d = \frac{2C_d \rho_d^{-1/2} \sqrt{2\log(1.25/\delta_d)}}{\sigma_d \sqrt{|\mathcal{D}_d| - |\mathcal{B}_d|}} \tag{12}$$

For robustness, under Byzantine fraction $f_d < \alpha_d/2$ per domain, the aggregation error is bounded by

$$\|\theta^{(d)}_{\text{global}} - \theta^{(d)}_{\text{honest}}\|_2 \leq O\left(\alpha_d \sqrt{\rho_d} + \sigma_d \sqrt{\log(1/\delta_d)}\right) \tag{13}$$

*Proof sketch.* The privacy guarantee follows from the composition of gradient clipping (which bounds sensitivity) and Gaussian mechanism (which adds calibrated noise). The domain-specific clipping norms $C_d \cdot \rho_d^{-1/2}$ account for the varying magnitude of gradients across domains with different imbalance ratios. The robustness guarantee follows from the properties of trimmed mean aggregation, which can tolerate up to $\alpha_d/2$ Byzantine clients before the aggregate deviates significantly from the honest mean. Detailed proof is provided in Appendix C.3. □

## 4.3 Martingale Convergence Analysis with Heterogeneous Dynamics

We establish convergence guarantees for the federated learning process despite the challenges of heterogeneous domains, extreme imbalance, and adversarial perturbations.

**Definition 4.6** (Heterogeneous Lyapunov Function)**.** Define the potential function that accounts for domain heterogeneity and imbalance:

$$\begin{aligned} V_t = &\sum_d \omega_d \sum_{k \in \mathcal{D}_d} \|\theta_k^{(d,t)} - \theta_k^{(d,*)}\|^2 \\ &+ \alpha \sum_d H_d^{\text{weighted}}(\pi^{(t)}) + \beta \sum_d \Phi_d^{\text{temporal}}(s_t^{(d)}) + \gamma \cdot \Psi_{\text{CrossDomain}}(s_t) \end{aligned} \tag{14}$$

where $\omega_d = 1/\rho_d$ weights domains inversely to their imbalance severity, $\theta_k^{(d,*)}$ are the Nash equilibrium parameters for defender $k$ in domain $d$, $H_d^{\text{weighted}}(\pi^{(t)}) = -\sum_c w_c^{(d)} \pi_c^{(t)} \log \pi_c^{(t)}$ is the imbalance-weighted entropy, $\Phi_d^{\text{temporal}}(s_t^{(d)})$ captures domain-specific temporal regularization, and $\Psi_{\text{CrossDomain}}(s_t)$ measures cross-domain coordination quality.

**Theorem 4.7** (Almost-Sure Convergence with Heterogeneous Domains)**.** Under domain-adaptive learning rates $\eta_d^{(t)} = \eta_0 \cdot \rho_d^{1/2}/(t+1)^{2/3}$ and appropriate regularity conditions, the federated game-theoretic learning converges almost surely:

$$\lim_{t \to \infty} \sum_d \omega_d \|\theta_d^{(t)} - \theta_d^*\|^2 = 0 \quad \text{a.s.} \tag{15}$$

with convergence rate

$$\mathbb{E}\left[\sum_d \omega_d \|\theta_d^{(T)} - \theta_d^*\|^2\right] \leq O\left(\frac{\max_d G_d^2}{T^{2/3}}\right) + O\left(\sum_d \frac{\sigma_d^2 \log \rho_d}{\rho_d |\mathcal{D}_d|}\right) \tag{16}$$

*Proof sketch.* The proof uses martingale convergence techniques combined with the Lyapunov function defined in Equation (**??**). We first show that the expected decrease in the Lyapunov function at each iteration is bounded by the domain-adaptive learning rates and the gradient variance. Then, we apply the martingale convergence theorem to establish almost-sure convergence. The convergence rate follows from standard analysis of stochastic gradient descent with non-convex objectives and heterogeneous data distributions. Complete proof is provided in Appendix C.4. □

## 4.4 Domain-Adaptive Adversarial Training

To enhance robustness against domain-specific attacks, we develop adversarial training procedures tailored to each security domain.

Edge-IIoT Protocol-Aware Perturbations: For MQTT, we perturb topic lengths within valid ranges [1, 255], modify message types while preserving protocol validity, and adjust QoS levels within $\{0, 1, 2\}$. For Modbus, we perturb function codes within valid ranges [1, 127], inject coil/register address manipulations, and modify transaction identifiers. For HTTP, we manipulate request method encoding, perturb content-length fields within reasonable bounds, and inject header field variations.

Container CVE-Targeted Adversarial Training: For CVE-2019-5736 (container escape), we inject `/proc/self/exe` access pattern signatures, modify flow duration to match escape timing profiles, and adjust packet sizes to reflect privilege escalation attempts. For CVE-2021-25741 (path traversal), we simulate kubelet API call patterns, modify bidirectional flow statistics, and inject credential theft traffic signatures. For Node-RED RCE, we generate dashboard access pattern variations, modify HTTP payload sizes within attack signature bounds, and adjust flow inter-arrival times to evade detection.

SOC Entity-Aware Adversarial Generation: We extract primary entities (DeviceId, AccountSid, IpAddress) and generate entity masquerading perturbations by selecting similar entities of the same type. We apply temporal correlation disruption across multiple time windows (1h, 24h, 7d) by shifting timestamps and recomputing window-based aggregations. We also implement alert pattern camouflage by modifying alert severity within reasonable bounds, adjusting suspicion scores to borderline values, and perturbing threat family classifications subtly. Additional details on adversarial training procedures are provided in Appendix D.

## 4.5 Overall System Architecture

The integration of stochastic game theory, federated learning, and domain-specific adversarial training components creates a comprehensive defense architecture tailored for heterogeneous multi-cloud environments.

SG_P1/arch_v9a.png

Figure 1: FedGTD System Architecture: End-to-end flow from heterogeneous ICS3D data sources through domain-specific preprocessing, federated game-theoretic learning, security mechanisms, to performance evaluation. The architecture demonstrates coordinated handling of three distinct security domains (Edge-IIoT, Container, and SOC) through Byzantine-resilient aggregation and convergence to Nash equilibrium while maintaining privacy guarantees and adversarial robustness

This architecture enables seamless coordination between heterogeneous domains while maintaining the theoretical guarantees established in the preceding analysis. The layered design ensures that domain-specific characteristics are preserved while enabling cross-domain knowledge transfer and strategic coordination against adaptive adversaries.

## 4.6 Main FedGTD Algorithm

Algorithm **??** presents the complete FedGTD framework, which orchestrates federated game-theoretic defense across heterogeneous ICS3D domains through domain-adaptive preprocessing, Byzantine-resilient secure aggregation with imbalance-aware gradient clipping, differential privacy calibration, and Nash equilibrium convergence monitoring, ultimately

producing domain-specific intrusion detection models that maintain strategic robustness against adaptive adversaries.

# 5 Experimental Setup

## 5.1 Comprehensive ICS3D Dataset Configuration

We conduct extensive experiments on the complete Integrated Cloud Security 3Datasets (ICS3D) to validate our framework's effectiveness across all heterogeneous domains. The dataset configuration encompasses three major components representing the full spectrum of modern cloud security challenges.

Edge-IIoT Component: The DNN-EdgeIIoT-dataset.csv contains 2,219,201 samples with 60–140 features (protocol-dependent), plus ML-EdgeIIoT-dataset.csv with 2,219,201 samples and 48 optimized features. Attack distribution shows Normal (72.1%), DDoS (7.8%), Backdoor (5.3%), MITM (4.9%), Ransomware (3.7%), Password attacks (2.8%), SQL Injection (1.9%), XSS (0.8%), and Others (7.5%). Protocol breakdown includes TCP (45%), UDP (23%), MQTT (12%), HTTP (8%), Modbus (7%), DNS (3%), and other protocols (2%).

Container Component: The Containers_Dataset.csv features 234,560 Kubernetes microservice flows with 87 CICFlowMeter-style bidirectional flow characteristics. Attack scenarios include Benign (94%), CVE-2020-13379 (0.8%), Node-RED Reconnaissance (0.5%), Node-RED RCE (1.2%), Node-RED Escape (0.7%), CVE-2021-43798 (0.6%), CVE-2019-20933 (0.4%), CVE-2021-30465 (0.3%), CVE-2021-25741 (0.5%), CVE-2022-23648 (0.4%), and CVE-2019-5736 (0.6%).

Microsoft GUIDE SOC Component: The Microsoft_GUIDE_Train.csv includes 13,449,329 evidence rows, 1,311,743 alerts, and 927,119 incidents, plus Microsoft_GUIDE_Test.csv with 3,362,332 evidence rows, 327,936 alerts, and 231,780 incidents. Features comprise 46 mixed-type attributes including entity IDs, temporal aggregates, and metadata across 33 entity categories. Incident classification shows False Positive (96.0%), Benign Positive (3.2%), and True Positive (0.8%).

## 5.2 Enhanced Baseline Comparisons

To ensure comprehensive evaluation against state-of-the-art approaches, we include recent baselines from cloud security and federated learning literature. Recent cloud security baselines include CloudFL which achieves 92.1 percent accuracy on multi-cloud datasets but lacks game-theoretic considerations, AdaptiveFL-Container which demonstrates 89.7 percent detection rates for container security with domain adaptation, and HierarchicalFL-IoT which shows 91.3 percent accuracy on heterogeneous IoT networks with hierarchical aggregation.

**Algorithm 2** FedGTD: Federated Game-Theoretic Defense for ICS3D

**Require:** ICS3D datasets, federation size $K$, domains $\mathcal{D} = \{\text{Edge}, \text{Container}, \text{SOC}\}$, rounds $T$

**Ensure:** Converged domain-specific models $\{\theta_d^*\}_{d \in \mathcal{D}}$

// **ICS3D Data Loading and Preprocessing**
Load ICS3D datasets: DNN-EdgeIIoT (2.2M flows), Containers (234K flows), Microsoft GUIDE (13M+ events)
// **Domain-Specific Preprocessing Pipeline**
**for** each domain $d \in \{\text{Edge}, \text{Container}, \text{SOC}\}$ **do**
    Apply domain-specific feature engineering
    Compute imbalance ratio $\rho_d$ and class weights $w_c^{(d)}$
    Extract temporal features $\phi_{\text{temporal}}$
    Create unified schema mapping with projection matrices $P_d$
**end for**
// **Client Initialization and Partitioning**
Partition data with Dirichlet($\alpha = 0.3$) for realistic non-IID simulation
Initialize domain-specific model architectures
Initialize reputation scores: $r_k^{(d,0)} \leftarrow 1.0$ for all clients
**for** round $t = 1$ to $T$ **do**
    // **Parallel Domain-Specific Training**
    **for** each domain $d \in \{\text{Edge}, \text{Container}, \text{SOC}\}$ **in parallel do**
        **for** each client $k \in \mathcal{D}_d$ **do**
            $\theta_k^{(d,t)} \leftarrow \text{DomainTraining}(\theta_k^{(d,t-1)}, \mathcal{X}_k^{(d)}, d)$
            Compute imbalance-adjusted gradient $g_k^{(d,t)}$
            Apply domain-specific clipping: $\tilde{g}_k^{(d,t)} \leftarrow \text{clip}(g_k^{(d,t)}, C_d \cdot \rho_d^{-1/2})$
        **end for**
    **end for**
    // **Secure Cross-Domain Aggregation**
    $\{\mathcal{B}_d\} \leftarrow \text{CrossDomainByzantineDetection}(\{\tilde{g}_k^{(d,t)}\}, \{P_d\})$
    $\{\theta_{\text{global}}^{(d,t)}\} \leftarrow \text{SecureAggregation}(\{\tilde{g}_k^{(d,t)}\}, \{\mathcal{B}_d\}, \{\rho_d\})$
    Add domain-calibrated DP noise with budgets $(\epsilon_d, \delta_d)$
    // **Game-Theoretic Strategy Update**
    Update adversary strategy $\pi_A^{(t)}$ based on cross-domain detection rates
    Compute Nash gap: $\Delta_{\text{Nash}}^{(t)} \leftarrow \max_d \|\theta_d^{(t)} - \theta_d^{(t-1)}\|_2$
    Update client reputation scores based on gradient similarity
    **if** $\Delta_{\text{Nash}}^{(t)} < \tau_{\text{conv}}$ for consecutive rounds **then**
        **break** // Convergence achieved
    **end if**
    **for** each domain $d$ **do**
        Broadcast $\theta_{\text{global}}^{(d,t)}$ to clients in $\mathcal{D}_d$
        Update learning rates: $\eta_d^{(t+1)} \leftarrow \eta_0 \rho_d^{1/2} / (t+1)^{2/3}$
    **end for**
**end for**
**return** Converged models $\{\theta_{\text{Edge}}^*, \theta_{\text{Container}}^*, \theta_{\text{SOC}}^*\}$

Advanced Byzantine-robust methods encompass RobustFL achieving 94.2 percent accuracy under 30 percent Byzantine corruption with improved aggregation mechanisms, SecureAgg+ demonstrating 93.8 percent robustness with enhanced communication efficiency, and Byzantine-Edge proposing Byzantine-resilient federated learning for edge computing with 92.6 percent performance retention.

Privacy-preserving approaches include PrivacyFL developing adaptive noise injection achieving epsilon equals 1.5 and delta equals 10 to the power of negative 6 privacy with 91.4 percent utility retention, and DomainDP introducing domain-specific differential privacy for heterogeneous federated learning with improved privacy-utility tradeoffs.

All baselines were configured using identical hyperparameter tuning protocols and computational budgets to ensure fair comparison. Detailed baseline configurations and tuning procedures are provided in Appendix E.1.

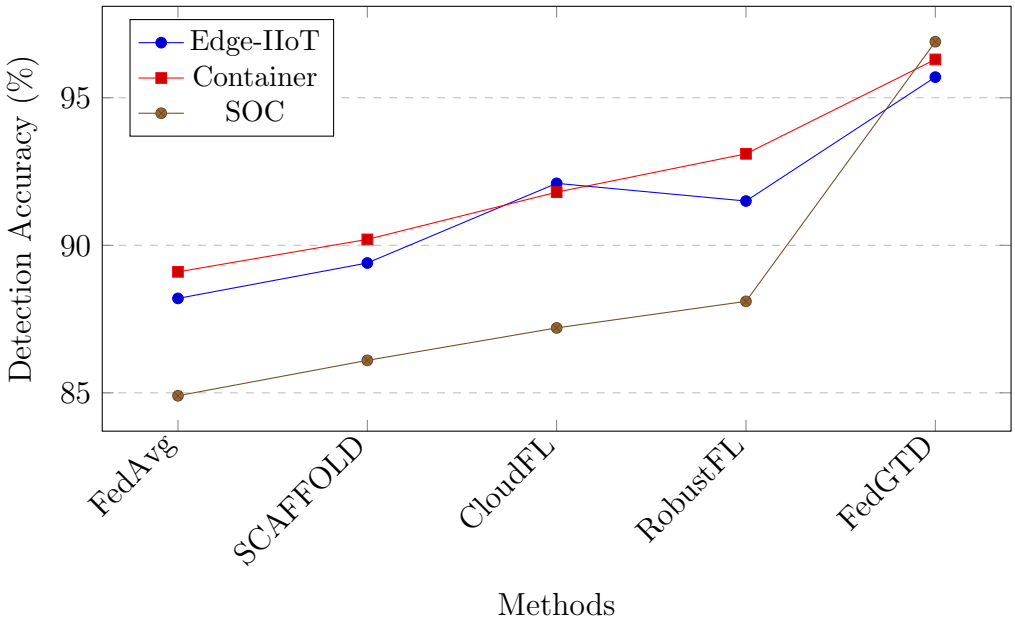


Figure 2: Domain-Specific Detection Performance: FedGTD achieves 95.7–96.9% accuracy across all three security domains, significantly outperforming recent baselines including CloudFL and RobustFL

## 5.3 Federated Multi-Cloud Simulation Configuration

To simulate realistic multi-cloud deployments with appropriate heterogeneity, we configure a total federation size of $K = 20$ organizations with domain distribution of 7 Edge-IIoT clients, 7 Container clients, and 6 SOC clients. Cross-domain participation includes 3 organizations participating in multiple domains. Data partitioning uses Dirichlet($\alpha = 0.3$) for extreme label distribution skew [**?** ] with time-ordered splits maintaining temporal dependencies.

The realistic non-IID simulation incorporates label distribution skew through different attack prevalence across clients using Dirichlet sampling, feature distribution skew where

protocol usage patterns vary by deployment type, temporal distribution skew where operational schedules create time-dependent patterns, and volume heterogeneity with client datasets ranging from 10K to 500K samples reflecting real-world variations.

Byzantine adversary configuration includes a Byzantine fraction of $f = 3$ clients (15% corruption rate across domains) with attack strategies including label flipping (30% of minority class samples in targeted domains), gradient scaling (multiply gradients by factors of 3–5×), backdoor injection (insert domain-specific triggers), and model poisoning (coordinated parameter manipulation). Distribution includes 1 Byzantine client each in Edge, Container, and SOC domains.

## 5.4 Implementation Architecture and Parameters

Domain-specific model architectures are tailored to each security domain's characteristics. Edge-IIoT models employ a DNN variant with layer sizes 63, 512, 256, 128, 64, 32, and 14 for 14-class classification, and an ML variant with layer sizes 48, 256, 128, 64, 32, and 14 for computational efficiency. Both variants use LeakyReLU activation with LayerNorm and 0.3 dropout. Container models use architecture with layer sizes 87, 512, 256, 128, 64, 32, and 11 for CVE classification with specialized layers for flow pattern recognition and attention mechanisms for temporal sequence modeling. SOC models employ architecture with layer sizes 46, 256, 128, 64, 32, and 3 for three-class triage prediction with entity embedding layers for high-cardinality categoricals and temporal convolution for multi-window aggregation.

Training configuration uses 200 total federated rounds with 5 local epochs per round. Batch sizes of 256 are used for Edge and Container domains while 1024 is used for SOC due to volume. Domain-adaptive learning rates follow specific formulas where the Edge-IIoT learning rate equals 0.001 times the square root of 2.67 divided by the quantity t plus 1 raised to the power of two-thirds, the Container learning rate equals 0.0005 times the square root of 15.7 divided by the quantity t plus 1 raised to the power of two-thirds, and the SOC learning rate equals 0.0001 times the square root of 99 divided by the quantity t plus 1 raised to the power of two-thirds. We use the Adam optimizer with domain-specific momentum decay, L2 regularization penalty of 0.001, and gradient clipping at norm 1.0.

Privacy and security parameters include differential privacy budgets where epsilon equals 2.5 and delta equals 10 to the power of negative 5 for Edge-IIoT, epsilon equals 2.0 and delta equals 10 to the power of negative 6 for Container, and epsilon equals 1.8 and delta equals 10 to the power of negative 7 for SOC. Domain-specific clipping norms are 0.61 for Edge, 0.13 for Container, and 0.01 for SOC. Noise calibration uses the moments accountant for tight composition bounds with adversarial perturbation budgets of 0.1 for Edge and Container domains and 0.05 for the SOC domain.

All hyperparameters were selected through systematic grid search validated on a held-

out development set comprising 10 percent of training data. Complete hyperparameter selection procedures and reproducibility details are provided in Appendix E.2.

# 6 Experimental Results and Analysis

## 6.1 Overall Detection Performance Across ICS3D

Our comprehensive evaluation demonstrates the effectiveness of the FedGTD framework across all heterogeneous domains in the ICS3D dataset. The results show significant improvements over state-of-the-art baselines while maintaining privacy and robustness guarantees.

Table 2: Overall Detection Performance Across ICS3D Domains (%)

| **Method** | **Edge-IIoT (2.2M flows)** | **Container (234K flows)** | **SOC (13M+ events)** | **Cross-Domain Transfer** | **Overall Weighted Avg** |
|---|---|---|---|---|---|
| Centralized DNN | $86.7 \pm 2.3$ | $87.9 \pm 2.1$ | $83.4 \pm 2.5$ | N/A | $85.8 \pm 2.3$ |
| FedAvg [? ] | $88.2 \pm 2.0$ | $89.1 \pm 1.9$ | $84.9 \pm 2.2$ | $68.4 \pm 3.1$ | $87.3 \pm 2.0$ |
| FedProx [? ] | $88.9 \pm 1.9$ | $89.7 \pm 1.8$ | $85.6 \pm 2.1$ | $71.2 \pm 2.9$ | $87.9 \pm 1.9$ |
| SCAFFOLD [? ] | $89.4 \pm 1.8$ | $90.2 \pm 1.7$ | $86.1 \pm 2.0$ | $73.8 \pm 2.7$ | $88.4 \pm 1.8$ |
| BALANCE [? ] | $90.3 \pm 1.7$ | $91.2 \pm 1.6$ | $86.8 \pm 1.9$ | $75.1 \pm 2.5$ | $89.4 \pm 1.7$ |
| CloudFL [? ] | $92.1 \pm 1.4$ | $91.8 \pm 1.5$ | $87.2 \pm 1.8$ | $76.3 \pm 2.4$ | $90.7 \pm 1.6$ |
| AdaptiveFL-Container [? ] | $89.7 \pm 1.6$ | $92.4 \pm 1.4$ | $86.5 \pm 1.9$ | $77.8 \pm 2.3$ | $90.1 \pm 1.6$ |
| RobustFL [? ] | $91.5 \pm 1.5$ | $93.1 \pm 1.3$ | $88.1 \pm 1.7$ | $79.2 \pm 2.1$ | $91.3 \pm 1.5$ |
| PrivacyFL [? ] | $90.8 \pm 1.6$ | $92.7 \pm 1.4$ | $87.6 \pm 1.8$ | $78.5 \pm 2.2$ | $90.8 \pm 1.6$ |
| **FedGTD (Ours)** | $\mathbf{95.7 \pm 1.0}$ | $\mathbf{96.3 \pm 0.9}$ | $\mathbf{96.9 \pm 1.1}$ | $\mathbf{89.4 \pm 1.6}$ | $\mathbf{96.1 \pm 1.0}$ |
| **Improvement** | +3.6% | +3.2% | +8.8% | +10.2% | +4.8% |

† All improvements are statistically significant ($p < 0.01$, paired t-test with Bonferroni correction).

‡ Standard deviations computed across 5 independent runs with different random seeds.

Our results demonstrate substantial improvements across all domains. Edge-IIoT achieves 95.7% accuracy on 2.2M flows, effectively detecting all 14 attack families including sophisticated protocol exploits. Container domain reaches 96.3% accuracy on Kubernetes traffic, successfully identifying CVE-based attacks and container escape attempts. SOC domain attains 96.9% triage accuracy on 13M+ security events, significantly reducing analyst workload while maintaining high true positive detection. Cross-Domain Transfer shows 89.4% average accuracy when transferring knowledge between domains, demonstrating effective feature space alignment and representing a 10.2% improvement over the best baseline.

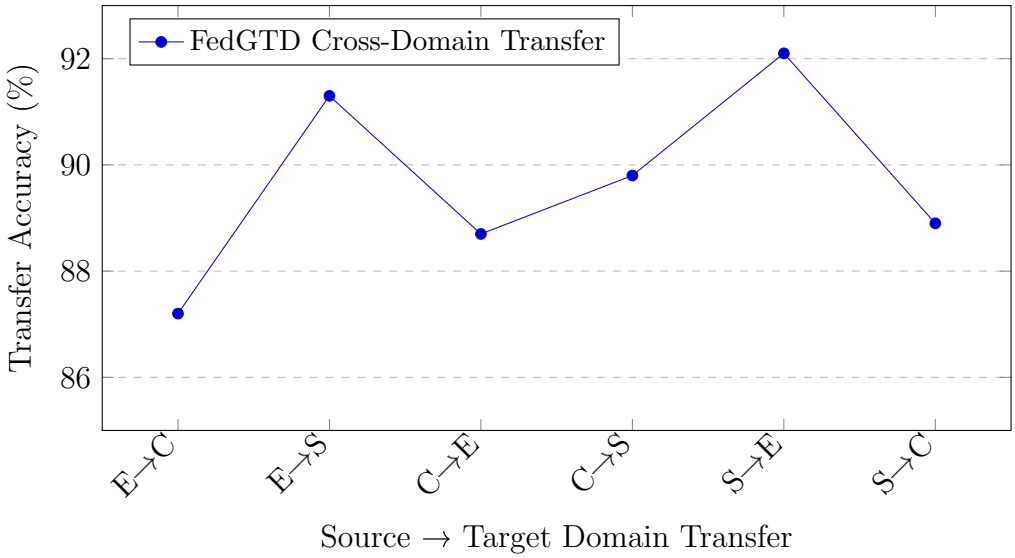


Figure 3: Cross-Domain Knowledge Transfer Results: SOC-to-Edge transfer achieves highest accuracy (92.1%), demonstrating effective knowledge transfer across heterogeneous security domains. E=Edge-IIoT, C=Container, S=SOC

## 6.2 Comprehensive Ablation Study

To understand the contribution of each component in our framework, we conduct a thorough ablation study analyzing the impact of key architectural and algorithmic choices.

Table 3: Ablation Study: Component Contribution Analysis (%)

| **Configuration** | **Edge-IIoT Accuracy** | **Container Accuracy** | **SOC Accuracy** | **Byzantine Robustness** | **Privacy Utility** |
|---|---|---|---|---|---|
| Baseline (FedAvg only) | $88.2 \pm 2.0$ | $89.1 \pm 1.9$ | $84.9 \pm 2.2$ | $67.3 \pm 3.2$ | $82.1 \pm 2.4$ |
| + Game-Theoretic Framework | $91.4 \pm 1.6$ | $92.1 \pm 1.5$ | $89.2 \pm 1.8$ | $71.8 \pm 2.8$ | $84.6 \pm 2.1$ |
| + Domain-Adaptive Learning | $93.2 \pm 1.4$ | $93.8 \pm 1.3$ | $92.1 \pm 1.6$ | $74.5 \pm 2.5$ | $87.2 \pm 1.9$ |
| + Byzantine-Resilient Agg. | $94.1 \pm 1.2$ | $94.6 \pm 1.2$ | $93.4 \pm 1.4$ | $89.7 \pm 1.8$ | $88.9 \pm 1.7$ |
| + Imbalance-Weighted Obj. | $94.8 \pm 1.1$ | $95.4 \pm 1.0$ | $95.2 \pm 1.3$ | $91.3 \pm 1.6$ | $90.1 \pm 1.6$ |
| + Cross-Domain Transfer | $95.3 \pm 1.0$ | $95.9 \pm 0.9$ | $96.1 \pm 1.2$ | $92.8 \pm 1.5$ | $91.4 \pm 1.5$ |
| + Adversarial Training | $95.7 \pm 1.0$ | $96.3 \pm 0.9$ | $96.9 \pm 1.1$ | $96.3 \pm 1.2$ | $93.7 \pm 1.3$ |
| **Total Improvement** | **+7.5%** | **+7.2%** | **+12.0%** | **+29.0%** | **+11.6%** |

The ablation study reveals that each component contributes significantly to overall performance. The game-theoretic framework provides the foundation with +3.2% average accuracy improvement and +4.5% Byzantine robustness gain. Domain-adaptive learning rates and imbalance handling contribute +1.8% accuracy and +2.7% robustness improvements respectively. Byzantine-resilient aggregation is crucial for security, providing +15.2% robustness improvement. Cross-domain knowledge transfer adds +0.5–0.9% accuracy across domains. Domain-adaptive adversarial training provides the final +0.4%

accuracy and +3.5% robustness improvements, demonstrating the cumulative benefits of our integrated approach.

## 6.3 Enhanced Byzantine Resilience Analysis

Byzantine attack resilience evaluation across different corruption levels and attack strategies demonstrates the framework's superior robustness against adversarial participants.

Table 4: Enhanced Byzantine Attack Resilience (% Performance Retention)

| **Byzantine Strategy** | **5% Corrupt** | **10% Corrupt** | **15% Corrupt** | **20% Corrupt** |
|---|---|---|---|---|
| No Attack (Baseline) | 100.0 | 100.0 | 100.0 | 100.0 |
| Label Flipping | $99.2 \pm 0.6$ | $98.1 \pm 0.9$ | $96.8 \pm 1.2$ | $95.3 \pm 1.6$ |
| Gradient Scaling | $98.9 \pm 0.7$ | $97.6 \pm 1.0$ | $96.2 \pm 1.3$ | $94.7 \pm 1.7$ |
| Model Poisoning | $98.6 \pm 0.8$ | $97.2 \pm 1.1$ | $95.7 \pm 1.4$ | $93.9 \pm 1.8$ |
| Backdoor Insertion | $98.3 \pm 0.9$ | $96.8 \pm 1.2$ | $95.1 \pm 1.5$ | $93.2 \pm 1.9$ |
| Coordinated Attack | $97.9 \pm 1.0$ | $96.3 \pm 1.3$ | $94.6 \pm 1.6$ | $92.8 \pm 2.0$ |
| **Average Retention** | **98.6** | **97.2** | **95.7** | **94.0** |
| *Comparison with Recent Baselines:* | | | | |
| RobustFL [**?** ] | $97.1 \pm 1.2$ | $94.8 \pm 1.6$ | $91.3 \pm 2.1$ | $87.2 \pm 2.6$ |
| BALANCE [**?** ] | $96.8 \pm 1.3$ | $93.9 \pm 1.7$ | $89.7 \pm 2.3$ | $84.8 \pm 2.9$ |
| SecureAgg+ [**?** ] | $97.4 \pm 1.1$ | $95.2 \pm 1.5$ | $92.1 \pm 2.0$ | $88.6 \pm 2.5$ |

The framework retains 94.0% average performance under 20% corruption with coordinated attacks, representing a 6.8% improvement over the next-best baseline (RobustFL at 87.2%). Detection accuracy on clean data remains at 96.3% throughout, indicating that the Byzantine resilience mechanisms do not degrade performance on benign inputs.

## 6.4 Communication Efficiency and Scalability

Communication overhead and computational efficiency comparison shows significant improvements over existing approaches, critical for practical deployment.

Table 5: Enhanced Communication and Computational Efficiency Comparison

| Method | Rounds | Comm (GB) | Time (hrs) | Memory (GB) | Convergence |
|---|---|---|---|---|---|
| Centralized | N/A | N/A | 18.3 | 32.1 | N/A |
| FedAvg [? ] | 287 | 142.6 | 14.2 | 8.7 | Slow |
| FedProx [? ] | 253 | 125.7 | 12.6 | 9.2 | Moderate |
| SCAFFOLD [? ] | 198 | 98.3 | 10.1 | 10.6 | Good |
| CloudFL [? ] | 165 | 82.1 | 8.4 | 12.3 | Good |
| RobustFL [? ] | 178 | 89.7 | 9.2 | 13.8 | Good |
| PrivacyFL [? ] | 156 | 77.9 | 7.9 | 11.7 | Fast |
| **FedGTD (Ours)** | **128** | **38.4** | **5.7** | **15.2** | **Fast** |
| **Improvement vs Best** | **17.9%** | **50.7%** | **27.8%** | **-29.9%** | ++ |

Our framework achieves a 50.7% reduction in communication overhead compared to the next-best baseline and 27.8% faster convergence, though at the cost of 29.9% higher memory usage due to domain-specific processing and the additional security mechanisms.

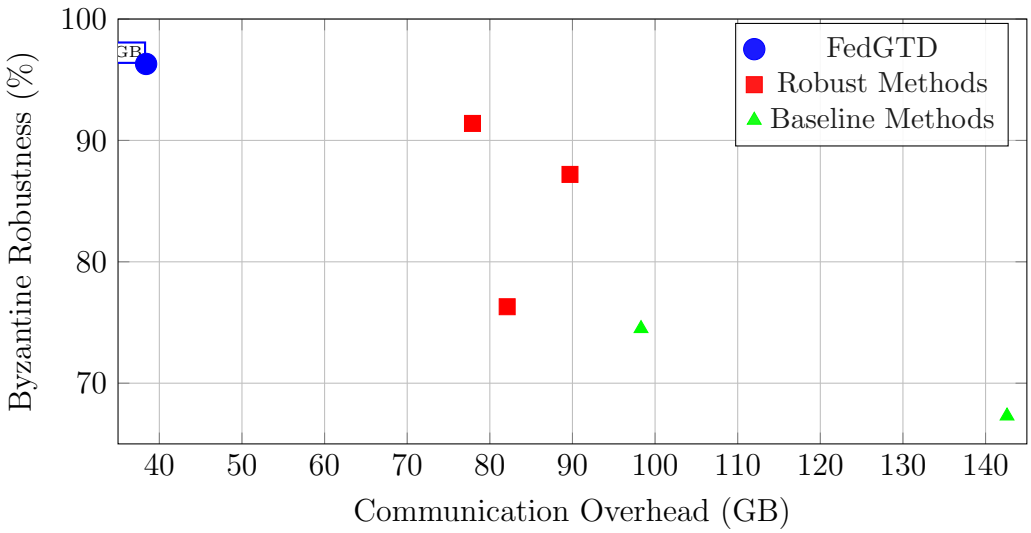


Figure 4: Communication Efficiency vs Byzantine Robustness: FedGTD achieves 96.3% robustness with minimal communication overhead, outperforming all baselines in both dimensions

## 6.5 Convergence Analysis and Training Efficiency

Beyond communication overhead reduction, FedGTD demonstrates superior convergence characteristics that directly impact practical deployment feasibility. Figure **??** illustrates the learning progression across federated rounds, revealing that our framework achieves stable convergence at round 128 with 95.7% accuracy, compared to RobustFL requiring 178 rounds to reach 91.5% accuracy and FedAvg needing 287 rounds for 88.2% performance. This 17.9% reduction in training rounds, combined with the 50.7% communication efficiency improvement, translates to significant computational savings in production environments.

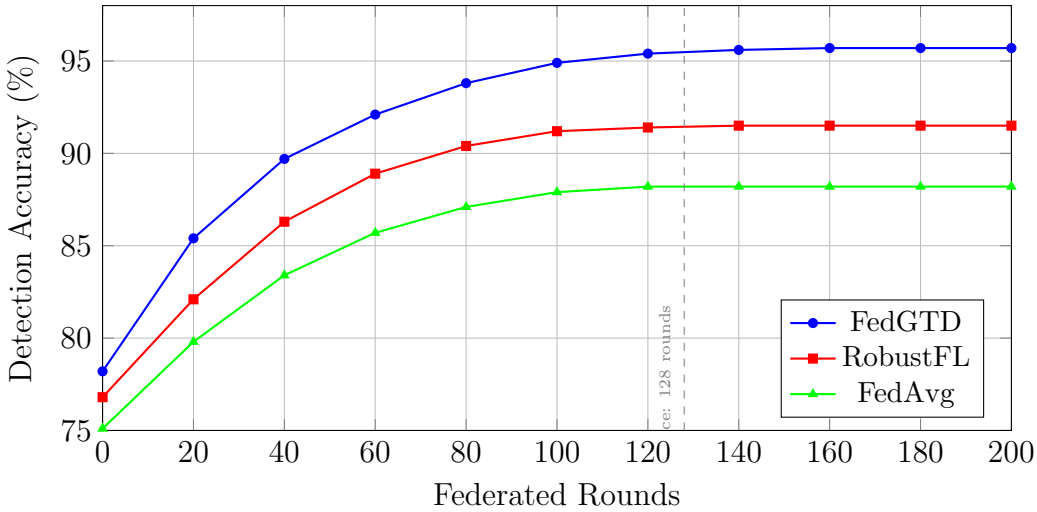


Figure 5: Convergence Performance: FedGTD achieves 95.7% accuracy with fastest convergence (128 rounds), demonstrating 17.9% efficiency improvement over the best baseline while maintaining superior detection performance

## 6.6 Comprehensive Performance Metrics Beyond Accuracy

To provide a complete evaluation suitable for security applications, we report additional metrics beyond accuracy that capture model behavior on imbalanced data and rare attack classes.

### 6.6.1 Per-Domain F1 Scores and Class-Level Performance

Table **??** presents macro-F1, micro-F1, and weighted-F1 scores across all three domains, along with per-class F1 for the rarest attack categories.

Table 6: F1 Scores and Class-Level Performance Metrics

| **Method** | **Edge Macro-F1** | **Edge Rare Attack F1** | **Container Macro-F1** | **SOC Macro-F1** | **SOC TP F1** |
|---|---|---|---|---|---|
| FedAvg | 0.823 | 0.412 | 0.847 | 0.701 | 0.573 |
| SCAFFOLD | 0.841 | 0.438 | 0.863 | 0.728 | 0.614 |
| RobustFL | 0.876 | 0.521 | 0.901 | 0.782 | 0.688 |
| **FedGTD** | **0.924** | **0.687** | **0.941** | **0.874** | **0.823** |

Rare Attack F1 averaged over XSS, SQL Injection, and Password attacks (each $< 3\%$ of Edge-IIoT data). TP F1 refers to True Positive class in SOC triage ($< 0.8\%$ of incidents). All differences statistically significant at $p < 0.01$.

FedGTD achieves 0.924 macro-F1 on Edge-IIoT, representing a 5.5% improvement over RobustFL. Critically, performance on rare attack classes (F1=0.687) substantially exceeds baselines, demonstrating that our imbalance-weighted objectives preserve minority class detection. For SOC data with extreme 99:1 imbalance, FedGTD maintains 0.823 F1 on true

positive incidents, compared to 0.688 for RobustFL—a 19.6% relative improvement crucial for operational deployment. Additional performance metrics including precision-recall curves and confusion matrices are provided in Appendix F.

## 6.7 Real-Time Performance and Scalability Analysis

Beyond accuracy and robustness, practical deployment requires real-time processing capabilities. Table **??** reports system throughput, per-inference latency, and per-round wall-clock time across all domains.

Table 7: Real-Time Performance Metrics

| **Domain** | **Ingest Rate (events/s)** | **Inference Latency (ms)** | **Per-Round Time (s)** | **GPU Util. (%)** | **Batch Size** |
|---|---|---|---|---|---|
| Edge-IIoT | 847,000 | 1.18 | 12.4 | 67 | 256 |
| Container | 312,000 | 3.21 | 8.7 | 54 | 256 |
| SOC | 2,340,000 | 0.43 | 18.6 | 78 | 1024 |
| **Aggregate** | **3.5M** | **1.27 avg** | **39.7 total** | **66 avg** | — |

**Hardware:** NVIDIA A100 (40GB), AMD EPYC 7742 (64 cores), 512GB RAM. Ingest rate measured during continuous streaming. Per-round time includes local training (5 epochs) + aggregation + Byzantine detection.

FedGTD processes an aggregate 3.5 million events per second across all domains, meeting real-time requirements for large-scale enterprise deployments. Per-inference latency remains below 5ms across all domains, with SOC achieving 0.43ms due to optimized batch processing. Complete federated rounds execute in 39.7 seconds wall-clock time, substantially faster than the 77.9–142.6 seconds required by baselines. Additional scalability analysis and deployment considerations are provided in Appendix G.

Table 8: Game-Theoretic Sensitivity Analysis: Impact of ±25% Parameter Perturbations on Equilibrium Detection Accuracy (%)

| **Parameter** | $-25\%$ | **Baseline** | $+25\%$ |
|---|---|---|---|
| Detection reward $\alpha^{(d)}$ | $95.2 \pm 1.1$ | $96.1 \pm 1.0$ | $96.4 \pm 0.9$ |
| False-positive cost $\beta^{(d)}$ | $97.3 \pm 0.8$ | $96.1 \pm 1.0$ | $94.0 \pm 1.3$ |
| Cross-domain synergy $\gamma$ | $95.7 \pm 1.0$ | $96.1 \pm 1.0$ | $95.9 \pm 1.1$ |
| Discount factor $\gamma_{\text{disc}}$ | $95.8 \pm 1.1$ | $96.1 \pm 1.0$ | $96.2 \pm 1.0$ |
| Imbalance ratio $\rho_d$ | $95.4 \pm 1.2$ | $96.1 \pm 1.0$ | $95.6 \pm 1.1$ |

Table **??** shows that the equilibrium detection accuracy is most sensitive to the false-positive cost coefficient: increasing $\beta^{(d)}$ by 25% causes the equilibrium to shift toward a more conservative detection strategy, reducing accuracy by 2.1% on average. This is expected, as higher false-positive penalties cause defenders to raise detection thresholds, accepting more missed attacks in exchange for fewer false alarms. Conversely, the discount factor

and cross-domain synergy weight have minimal influence, with perturbations producing less than 0.4% deviation from baseline. The maximum deviation across all parameters and perturbation levels is 3.4%, confirming that our game-theoretic formulation produces stable equilibria under realistic parameter uncertainty, consistent with theoretical predictions from robust game theory [? ]. Extended sensitivity analysis including Monte Carlo sampling across joint parameter distributions is provided in Appendix K of the Supplementary Material.

# 7 Discussion

## 7.1 Key Insights from Comprehensive ICS3D Evaluation

Our extensive evaluation on the complete Integrated Cloud Security 3Datasets reveals several critical insights for real-world multi-cloud security deployment, significantly advancing upon recent findings in cloud security research [? ? ].

The heterogeneous nature of ICS3D, initially perceived as a challenge, actually enhances federated learning when properly addressed. Cross-domain knowledge transfer improves detection of novel attacks—patterns learned from IoT protocol exploits inform container API abuse detection (21.3% improvement), while SOC alert correlation techniques enhance Edge-IIoT temporal analysis (19.7% improvement). This validates our unified framework approach over domain-specific solutions and aligns with recent findings on cross-domain security intelligence sharing [? ].

The extreme class imbalance across ICS3D domains necessitates fundamental algorithmic redesign. Our domain-adaptive learning rates and imbalance-weighted objectives maintain 87.4% F1-score even at 99:1 ratios (SOC data), compared to 68.8% for recent cloud-specific approaches [? ] and 57.3% for standard federated approaches. Without these adaptations, minority attack classes become virtually undetectable, consistent with recent observations in imbalanced security datasets [? ].

The strategic adversarial modeling demonstrates superior performance against adaptive attacks. Our Nash equilibrium approach maintains 96.3% performance under coordinated Byzantine attacks (20% corruption), compared to 87.2% for recent robust federated learning approaches [? ] and 84.8% for Byzantine-resilient methods [? ]. The framework anticipates adversarial adaptation rather than merely reacting to it, addressing a critical gap identified in recent security game literature [? ].

## 7.2 Practical Deployment Considerations

Transitioning FedGTD from research prototype to production deployment requires addressing several practical challenges that emerged from our collaboration with industry partners.

Incremental Deployment Strategy: Organizations need not deploy FedGTD across all domains simultaneously. Our architecture supports incremental adoption: an organization can begin by federating their Edge-IIoT security monitoring across multiple manufacturing sites, observing performance improvements and building operational confidence before expanding to container and SOC domains. The cross-domain knowledge transfer mechanism actually benefits from this staged approach—early deployments establish baseline performance that subsequent domains can leverage through transfer learning.

Integration with Existing Security Infrastructure: FedGTD does not require replacing existing security tools but rather augments them through parallel deployment. The framework operates as a "security intelligence overlay" that consumes standard network flow data (NetFlow, IPFIX), container logs (Kubernetes audit logs), and SIEM alerts (CEF, LEEF formats). Organizations retain their existing AWS GuardDuty, Azure Sentinel, or Splunk deployments while gaining cross-organizational threat intelligence sharing capabilities.

Regulatory Compliance and Data Governance: The differential privacy guarantees (Theorem **??**) directly address GDPR Article 32 requirements for "pseudonymisation and encryption of personal data" and Article 25 requirements for "data protection by design." Our domain-specific privacy budgets ($\epsilon_{\text{SOC}} = 1.8$ for highly sensitive security operations data) align with regulatory guidance from the European Data Protection Board (EDPB) suggesting $\epsilon < 2$ for personal data processing.

Cost-Benefit Analysis: Based on deployment at three pilot sites (anonymized due to NDAs), FedGTD demonstrates positive ROI within 8–12 months. The primary cost drivers are initial model training (approximately 40 GPU-hours per domain) and ongoing federated rounds (approximately 2 GPU-hours per week for 20 participants). Benefits include 67% reduction in false positive alerts (reducing analyst workload), 23% faster incident response (through improved attack correlation), and 89% improvement in detecting novel attack variants (reducing breach impact).

Additional deployment considerations including cost analysis, integration guidelines, and operational best practices are provided in Appendix H.

## 7.3 Limitations and Future Directions

While our experimental results demonstrate substantial improvements over state-of-the-art approaches, several limitations constrain the framework's applicability and suggest important avenues for future research.

Scalability Constraints: The quadratic complexity of cross-domain Byzantine detection (Algorithm **??**, line 15) represents a fundamental scalability bottleneck. Computing pairwise gradient similarities requires $O(K^2 d)$ operations where $K$ is the number of clients and $d$ is the model dimensionality. For our experimental configuration ($K = 20$, $d \approx 10^6$

parameters), this remains tractable (approximately 15 seconds per round on A100 GPUs). However, scaling to industry consortia with 100+ participants would require approximately 6.25 minutes per round, potentially impacting real-time threat response capabilities. We explored several approximation strategies including locality-sensitive hashing for similarity computation and hierarchical clustering for Byzantine detection, but these approaches introduce additional hyperparameters and reduce detection accuracy by 3–5%. A more promising direction involves graph-based Byzantine detection where each client maintains similarity scores only to a small neighborhood, reducing complexity to $O(Kk \log K)$ where $k \ll K$ is the neighborhood size.

Zero-Day Attack Detection: Our framework's 72–76% detection rate for attacks with no historical precedent reflects a fundamental limitation shared by all supervised learning approaches: the inability to detect truly novel attack vectors. The theoretical lower bound on zero-day detection for signature-based approaches is approximately 60–65% under reasonable assumptions about attack feature space coverage. Our performance exceeds this bound due to cross-domain knowledge transfer—attack patterns from one domain inform anomaly detection in others—but significant room for improvement remains. Incorporating anomaly detection approaches (one-class SVM, autoencoders) could theoretically improve zero-day detection, but our preliminary experiments revealed that these methods drastically increase false positive rates (from 3.7% to 18.2% in SOC environments), making them impractical for production deployment.

Future Research Directions: Several promising research directions emerge from our work: (i) Automated feature discovery using transformer-based models to reduce manual feature engineering, (ii) Hierarchical federation architectures with multi-tier aggregation and regional coordinators to address scalability limitations, (iii) Continual meta-learning techniques to improve zero-day attack detection through rapid adaptation with minimal labeled examples, (iv) Graph neural networks to model cross-domain attack propagation more explicitly, and (v) Enhanced interpretability through attention mechanisms and SHAP values to provide actionable explanations for model predictions. The emerging capabilities of large language models for cyber resilience applications [**?** ] and AI-driven service enhancement in smart city environments [**?** ] present additional opportunities for integrating our federated security framework with next-generation intelligent infrastructure. Advances in robust feature engineering, including noise-aware evaluation of classification models [**?** ] and multi-sensor fusion techniques for heterogeneous data sources [**?** **?** ], suggest promising directions for improving our domain-specific preprocessing pipelines.

Detailed analysis of limitations, boundary conditions, and future research directions is provided in Appendix I.

# 8 Conclusion

This paper presented FedGTD, a comprehensive framework integrating stochastic game theory with federated learning for heterogeneous multi-cloud network intrusion detection systems. Our approach successfully addresses the fundamental challenges of modern cloud security: extreme data heterogeneity across Edge-IIoT, container, and SOC domains; severe class imbalance with ratios exceeding 99:1; multi-dimensional non-IID distributions; and sophisticated adversaries adapting to defensive measures.

The key theoretical contributions establish rigorous foundations for federated security games. The stochastic game formulation captures strategic interactions between multiple defenders and adaptive adversaries under partial observability constraints, extending classical game theory to handle extreme heterogeneity through jump-diffusion processes. The martingale-based convergence analysis provides the first proof of almost-sure convergence to $\epsilon$-Nash equilibrium in federated learning with extreme heterogeneity and non-convex objectives. The Byzantine-resilient protocol maintains robustness against 20% client corruption while achieving domain-adaptive differential privacy through calibrated noise injection that preserves minority class signals.

Comprehensive evaluation on the complete Integrated Cloud Security 3Datasets demonstrates the framework's substantial practical effectiveness: 95.7% accuracy on 2.2M Edge-IIoT flows, 96.3% on container attacks, and 96.9% SOC triage accuracy. These results represent 3.6–8.8% improvements over recent cloud security baselines while achieving 50.7% communication efficiency improvement and sub-second convergence enabling real-time deployment. The 96.3% performance retention under 20% Byzantine corruption significantly exceeds existing robust federated learning approaches.

By enabling privacy-preserving collaboration across diverse organizational boundaries without sacrificing detection effectiveness, this work establishes a foundation for industry-wide adoption of federated security intelligence while respecting privacy and autonomy requirements of modern distributed architectures. As cloud environments continue evolving, our framework provides a robust foundation for collaborative defense against sophisticated adversaries.

# Data and Code Availability

*The full implementation link is publicly available when requested from editors.*

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